Microchemical Journal 135 (2017) 163–170

Contents lists available at ScienceDirect

Microchemical Journal

journal homepage: www.elsevier.com/locate/microc

A multivariate approach to the study of orichalcum ingots from the underwater Gela's archaeological site

Eugenio Caponetti [a,b], Francesco Armetta [a,*], Lorenzo Brusca [c], Delia Chillura Martino [a,b], Maria Luisa Saladino [a], Stefano Ridolfi [d], Gabriella Chirco [d], Mario Berrettoni [e,*], Paolo Conti [f], Nicolò Bruno [g], Sebastiano Tusa [g]

[a] Dipartimento Scienze e Tecnologie Biologiche, Chimiche e Farmaceutiche – STEBICEF, Università degli Studi di Palermo, Parco d'Orleans II, Viale delle Scienze pad. 17, I-90128 Palermo, Italy
[b] Centro Grandi Apparecchiature-ATeN Center, Università degli Studi di Palermo, Via F.Marini 14, I-90128 Palermo, Italy
[c] Istituto Nazionale di Geofisica e Vulcanologia – Sezione di Palermo, Via U. La Malfa 153, 90146 Palermo, Italy
[d] Labor Artis CR Diagnostica, Viale delle Scienze pad. 16, I-90128 Palermo, Italy
[e] Dipartimento di Chimica Industriale "Toso Montanari", UoS campus di Rimini, Università di Bologna, Viale dei Mille 29, I-40196 Rimini, Italy
[f] Scuola di Scienze e Tecnologie, Università di Camerino, Piazza dei Costanti, 62032 Camerino, Italy
[g] Soprintendenza del mare della Regione Siciliana, Palazzetto Mirto Via Lungarini 9, I-90133 Palermo, Italy

ARTICLE INFO

Article history:
Received 19 May 2017
Received in revised form 21 August 2017
Accepted 5 September 2017
Available online 6 September 2017

Keywords:
Orichalcum ingots
ICP-OES
ICP-MS
Chemometric approach

ABSTRACT

In this work a careful ICP-OES and ICP-MS investigation of 38 ancient ingots has been performed to determine both major components and trace elements content to find a correlation between the observed different features and the composition.
The ingots, recovered in an underwater archaeological site of various finds near Gela (CL, Italy), were previously investigated by X-Ray Fluorescence (XRF) spectroscopy to know the composition of the alloy and it was found that the major elements were copper and zinc, in a ratio compatible with the famous orichalcum similar to the contemporary brass that was considered a precious metal in ancient times. The discovery of huge amount this alloy is extraordinary.
Following a chemometric approach at first, the use of Principal Component Analysis (PCA) and Cluster Analysis (CA) allowed us to highlight three well-defined groups of ingots and to point out three ingots that appeared outlier with respect to the whole sample set. Linear Discriminant Analysis (LDA) and Soft Independent Modeling of Class Analogy (SIMCA) enabled us to confirm the difference between the hypothesized groups. The prediction power of the variables computed by SIMCA allowed us pointing out some elements able to differentiate each group.
The three well-defined groups of ingots resulting from the chemometric analysis were in agreement with the observations of some morphological parameters such as ingot shape, width, and length and weight and by the presence of different kind of patina.
The appearance of three distinctive families of ingots can indicate different geographical location of the furnace, different technology stages and/or different raw material used in melting process and the morphology is indicative of cast diverse technologies. These findings can signify the starting point for giving important insights in the archaeometric study of the orichalcum ingots regarding the provenience and the manufacture technologies.

1. Introduction

The present work concerns the deepening of knowledge about the treasure of 40 antique ingots recovered from the seabed near Gela in the southern coast of Sicily in 2015 [1]. In a previous paper, by analysing the contest of the discovery it has been done the assumption that the ingots are part of a precious cargo transported in an ancient ship that was wrecked near the Gela coast [2]. This ship was dated to the end of VI century BC according to the pottery found nearby. A photo of the context of the discovery is showed in Fig. 1.

As example the photo of five representative ingots is reported in Fig. 2.

Ingots morphology suggested a casting mono-valve mold production. The mold contact surface is poorly defined, indicating the use of a rough refractory material. It can be excluded that two ingots were prepared by using the same mold. The presence of ripples on the surface exposed to the air during the cooling process, consequence of a rapid solidification of the melted alloy, indicated that the temperature was near the melting temperature. The presence of numerous impurities among the ripples was attributed to a separation process that

* Corresponding authors.
E-mail address: francesco.armetta01@unipa.it (F. Armetta).

http://dx.doi.org/10.1016/j.microc.2017.09.003
0026-265X/

Fig. 1. Photo of the context of the discovery.

underwent during the fusion. This suggested that a recasting of already machined objects was not used in order to obtain the ingots, therefore a primary production of the alloy could be assumed. All these findings suggested a rudimentary mold production compatible with the age of the ship.

In the previous work a preliminary investigation on 38 ingots was performed in situ by using a portable Energy dispersive X-Ray Fluorescence (XRF) instrument in order to determine the composition of the alloy. Results evidenced that all the ingots are constitutes by an alloy whose major elements are copper and zinc. The weight ratio between the two metals, Cu/Zn, varies between 2.7 and 5.7; these values are compatible with the composition of the famous orichalcum, considered a precious metal in ancient times. This results supports the hypothesis that, according to the pottery found nearby, this precious cargo was transported in an ancient ship dated to the end of VI century BC. This finding, archaeologically speaking, is very important because, to our knowledge, it is the first time that orichalcum is found in the shape of ingots and in so large amount. In the Greek world, orichalcum was the rarest and most expensive alloy, the value was second after gold and silver. It had previously been found only in small items; however, Plato wrote that the walls of mythical Atlantis were plastered with orichalcum, a metal connected with the Gods, according to poets of archaic Greece. The production of orichalcum from the antiquity up to XVI century was realized by the so called cementation process. This consists in a series of chemical reactions in a more or less closed crucible at temperature of 1000 ± 100 °C. The Zn contained in zinc ores such as sphalerite (ZnS_2) or smithsonite ($ZnCO_3$) is firstly reduced by heating and subsequently the vapor of reduced Zn partly diffuse in the Cu to form the Cu-Zn alloy. The temperature should be confined between 917° (Zn vapors) and 1083° (melting point of Cu) [3,4].

Recently Fan and others [5] showed that malachite [$Cu_2(CO_3)(OH)_2$], calamite [Zinc carbonate $ZnCO_3$ or smithsonite and zinc silicate $Zn_4Si_2O_7(OH)_2 \cdot H_2O$ or hemimorphite.] and charcoal form brass when warmed in a closed device. The product composition depends on the temperature "When the temperature is 800 °C, the brass formed contains Zn less than 5%; when the temperature is 850 °C, the formed brass contains Zn more than 10%; when the temperature is over 900 °C, the Zn content of formed brass can exceed 20%". A little bit higher temperature is needed to obtain brass by fusion. Furthermore the raw materials can contain other minority minerals that are characteristic of the cave.

The composition of the alloy can be investigated by various techniques. The XRF spectrometry, already applied to the study of the ingots in the previous work [2] is a non-invasive and non-destructive technique, in addition it can be performed by portable instruments, but unfortunately it can give information only on the main elements [6].

Relevant information about the origin of the raw materials and the production techniques may be obtained by more sophisticated analyses of the ingots and of their trace elements [7]. For this purpose, it was important that the authorities perceived that such knowledge is worth of the sacrifice of few milligrams of orichalcum ingots. As the valuation of trace elements is very important for an accurate multi-elemental fingerprint of the sample, in order to obtain plentiful analytical data to be examined by means of chemometric methods [8,9], analysis of main and trace elements were performed by Inductively Coupled Plasma Optical Emission Spectrometry (ICP-OES) and Inductively Coupled Plasma Mass Spectrometry (ICP-MS) respectively. One of the advantages of the use these techniques is the higher detection power attained, resulting in much improved limits of quantification, which can be particularly relevant for elements present at trace levels in micro samples. Both ICP methods, require small sample amounts that were sampled by drilling a micro well on each ingot [10].

Other techniques could provide with the same or a better accuracy the concentration of the trace elements, but the main reason to use ICP-OES and ICP-MS techniques was the common presence of these instrumentations in the modern analytical laboratories, and the established use for the high quality measurements concerning metal samples [11,12]. Furthermore ICP techniques have been widely used to analyse ancient metal objects [13–15] thus the results obtained in

Fig. 2. Photo of 5 representative ingots.

this study could contribute to the constitution of a database for the orichalcum artefacts coming from the Mediterranean and Middle East regions, that in the future could be useful for the determination of the geographical distribution of ancient artefacts of the same alloy.

2. Materials and methods

2.1. Materials

Small amounts of alloy from each ingot were sampled by drilling a micro well by using the Dremel 400 apparatus. Approximately 20 mg of each sample was dissolved by using 3 mL of aqua *regia* that was prepared by mixing hydrochloric acid and nitric acid, (both from J.T. Baker Ultrex) in a volumetric ratio 3:1. Properly diluted solutions (1:101 in volume) were prepared with distilled water (18.2 MΩ·cm) obtained by a Milli-Q Integral 5 apparatus.

2.2. ICP-OES

The main elements of the alloy were determined by using an ICP-OES Jobin Yvon Ultima2 at wavelength of 324.754 nm for Cu and 213.856 nm for Zn (detection limits were 0.5 and 0.1 μg/L respectively). The instrument is equipped with a concentric nebulizer Meinhard, a cyclonic spray chamber, a three-channel peristaltic pump and an auto sampler. The calibration routine was performed on diluted mix of 2 single element stock solutions (1000 mg/L) with 6 calibration points prepared freshly in 10 mL polyethylene tubes by dilution with 1% nitric acid.

The element concentrations in the aqueous solutions were determined by using calibration solutions in a range 5–100 mg/L of the Cu and Zn elements and calculating the weighted regression curve built with 6 calibration points. Element contents in the analysed samples were calculated using the spectrometer software (ICP Analyst, version 5.4).

The sensitivity variations were monitored by Yttrium at 324.306 nm with 1 mg/L concentration as internal standard added directly online. The precision of the analysis, checked running 3 replicates, was always within ±5%. Data accuracy was evaluated by analysing standard reference materials (Spectrapure Standards WW1 and 2) with error <10%.

2.3. ICP-MS

The relative amounts of the trace elements were determined by ICP-MS Agilent 7500ce equipped with a Micromist nebulizer, a Scott double pass spray chamber, a three-channel peristaltic pump, an auto sampler (ASX-500, Cetac), and an octopole reaction system (ORS) to remove spectral interferences (detection limits in the order of 1 μg/L). The mass spectrometer was calibrated with a multi-element standard solution with 10 calibration points. The multi-elements standard solution was prepared by dilution from a single and a multi-element stock solution, at 1000 mg/L and 100 mg/L respectively. Multi-element stock calibration solutions were always freshly prepared in 10 mL polyethylene tubes by dilution using 1% nitric acid for trace analysis as diluent.

Table 1
Main component and trace element concentrations as well as the copper/zinc weight ratio obtained by ICP-OES and ICP-MS techniques.

Sample	%				Concentration in PPM																	w/w
	Cu	Zn	Pb	Fe	Ni	Ag	Sb	As	Co	Cd	Sn	Te	Bi	Mn	Li	Al	V	Cr	Rb	Sr	Ba	Cu/Zn
S1	73	21	5.60	0.30	580	146	55.7	9.85	2.79	12.2	9.70	0.57	41.4	19.9	5.55	11.5	3.71	3.17	0.08	1.00	0.19	3.5
S2	87	16	4.29	0.53	534	199	132	36.4	8.44	20.0	4.80	1.00	72.8	20.8	2.17	4.46	1.55	1.31	0.07	0.15	0.18	5.5
S3	73	28	5.91	0.78	451	131	105	23.8	8.47	16.5	6.70	0.64	29.0	15.7	4.00	8.89	3.33	2.70	0.14	8.90	0.35	2.6
S4	80	20	6.17	0.39	558	171	70.0	11.1	4.83	11.9	9.34	0.82	60.7	23.2	6.16	13.8	6.08	4.91	0.14	0.29	0.39	3.9
S5	78	22	3.87	0.55	513	156	67.0	14.5	6.49	5.30	7.69	0.58	58.7	23.3	3.87	11.6	4.38	3.69	0.12	0.91	0.45	3.5
S6	81	22	6.51	0.19	579	159	60.3	8.32	1.88	9.60	8.54	0.96	61.4	15.7	2.12	4.36	2.29	1.90	0.11	0.45	0.75	3.7
S7	75	20	6.45	0.33	559	130	107	211	3.37	11.6	17.6	0.80	51.0	22.3	1.75	5.50	2.16	1.57	0.13	0.34	0.50	3.8
S8	82	24	6.13	0.31	548	160	75.4	9.66	3.19	7.60	6.98	0.83	58.5	23.7	1.99	13.3	3.55	2.36	0.11	1.45	0.26	3.4
S9	80	18	4.63	0.44	482	159	109	24.2	5.28	30.5	3.92	0.91	91.3	23.7	1.69	8.40	2.70	1.78	0.07	0.11	0.43	4.4
S10	72	28	5.92	0.76	435	105	111	23.5	8.29	16.2	6.65	0.61	27.6	19.5	1.23	6.77	2.00	1.41	0.12	0.21	0.30	2.5
S11	76	19	5.93	0.43	525	138	53.2	15.1	5.13	8.50	8.29	0.86	54.5	12.9	0.92	6.57	1.71	1.28	0.10	0.56	0.24	4.1
S12	69	17	6.67	0.50	657	141	50.1	13.8	4.45	8.30	8.14	0.87	55.0	13.4	1.20	3.22	1.65	1.10	0.10	3.13	0.22	4.1
S13	78	20	3.30	1.06	458	189	549	197	1.60	17.3	38.3	1.00	181	86.6	1.55	9.28	3.08	2.23	0.07	0.52	1.17	4.0
S14	74	25	4.70	0.44	433	81.7	88.2	14.6	5.08	20.3	1.98	0.71	26.1	29.7	1.13	6.51	2.49	1.55	0.10	0.17	0.19	3.0
S15	84	20	5.20	0.14	549	171	107	20.6	1.06	23.4	4.24	0.95	92.0	12.0	0.74	3.44	1.58	1.08	0.08	0.08	0.17	4.1
S16	82	19	4.72	0.16	560	162	103	19.9	1.14	22.2	4.00	0.89	82.1	12.4	0.67	21.2	2.32	2.53	0.08	0.31	0.31	4.2
S17	77	20	6.92	0.79	500	177	153	54.7	9.06	31.8	65.8	0.81	27.9	5.8	0.40	3.31	0.93	0.67	0.08	7.27	0.35	3.8
S18	83	19	4.57	0.44	515	158	111	22.1	5.09	28.4	3.80	0.99	89.3	21.5	0.52	4.08	1.69	1.01	0.09	11.8	0.27	4.4
S19	71	28	5.25	0.72	494	114	101	38.1	6.14	18.5	10.4	0.72	33.6	17.7	1.49	8.36	2.61	1.54	0.16	4.89	3.37	2.6
S20	76	19	6.31	0.41	535	139	51.5	14.0	4.41	8.20	8.11	0.97	56.3	13.4	0.62	2.91	1.34	0.79	0.06	0.42	0.40	4.1
S21	74	21	4.88	0.49	423	80.0	68.0	14.4	5.67	22.5	1.99	0.91	25.3	26.3	0.42	2.49	0.97	0.59	0.07	0.70	0.23	3.5
S22	71	21	6.00	0.93	606	142	175	55.3	15.03	11.8	42.0	0.56	46.3	25.7	0.85	11.6	2.82	2.19	0.12	15.1	3.61	3.4
S23	72	22	6.18	0.33	623	121	67.2	9.12	2.61	11.6	9.07	0.58	40.1	18.3	0.73	5.38	2.52	1.45	0.11	0.08	0.14	3.3
S24	76	18	6.49	0.28	645	172	52.5	9.43	1.94	9.10	9.75	1.00	79.7	14.6	0.40	3.76	1.39	0.86	0.08	0.15	0.28	4.3
S25	78	18	1.05	0.75	496	162	32.2	56.7	0.34	5.00	4.10	0.91	49.8	29.5	0.61	5.31	2.02	1.27	0.09	0.17	0.32	4.3
S26	75	20	6.35	0.45	543	137	57.1	9.50	4.95	11.2	8.29	0.85	58.6	23.4	0.49	14.6	1.69	1.10	0.07	0.19	0.25	3.8
S27	72	20	6.48	0.41	558	140	54.5	9.10	4.34	10.6	8.19	0.89	57.0	18.4	0.36	3.05	1.49	0.85	0.07	6.73	0.26	3.7
S28	76	21	5.91	0.83	471	172	139	46.5	8.88	54.7	1.89	0.66	27.0	12.5	0.41	6.32	2.03	1.60	0.09	4.00	4.59	3.6
S29	69	28	6.57	0.81	492	121	122	26.4	7.79	18.8	9.63	0.71	33.3	22.1	0.53	5.25	2.74	1.81	0.13	0.93	0.38	2.5
S30	86	16	4.61	0.57	467	196	158	37.7	10.4	19.8	4.90	0.99	74.2	22.5	0.19	2.16	1.13	0.67	0.07	0.28	0.20	5.4
S31	77	25	5.25	0.73	400	89	101	18.4	9.98	22.6	2.89	0.68	26.3	32.6	0.31	3.96	1.80	1.05	0.10	0.14	0.24	3.1
S32	82	22	7.22	0.72	461	148	148	42.8	9.05	53.8	2.84	0.95	39.1	14.0	0.39	4.74	2.51	1.55	0.12	0.18	0.31	3.7
S33	82	20	5.14	0.45	492	166	117	22.8	5.26	28.5	3.59	1.00	87.4	21.8	0.14	2.02	1.00	0.59	0.06	0.33	0.16	4.2
S34	80	16	4.48	0.47	468	187	132	33.8	7.64	18.8	4.52	1.02	70.4	18.7	0.14	2.18	0.89	0.57	0.06	0.09	0.19	5.0
S35	78	19	6.62	0.24	539	174	57.4	9.02	1.91	8.9	9.78	0.99	80.3	14.4	0.17	2.63	1.31	0.72	0.08	0.24	0.17	4.2
S36	78	20	7.71	0.18	683	175	64.8	7.04	2.36	6.4	6.57	1.00	62.0	14.2	0.30	5.24	2.19	1.97	0.11	0.42	0.23	3.9
S37	75	23	6.63	0.23	464	131	66.0	7.51	3.35	9.7	5.62	0.65	40.6	15.2	0.10	2.68	1.23	1.02	0.09	2.88	0.12	3.2
S38	90	21	2.33	0.38	656	165	125	201	0.90	24.8	15.7	1.75	151	79.1	0.40	8.97	3.95	2.34	0.22	2.20	0.42	4.3

The error is on the last digit.

The sensitivity variations were monitored using ^{103}Rh, ^{115}In, and ^{185}Re internal standards added directly online by an appropriate device that mixes the internal standard solution into the sample just before the nebulizer. The concentration of the internal standard was about 10 μg/L in all sample and calibration solutions. The precision of the analysis, checked running five replicates of every standard and sample, was always within $\pm$10%. Data accuracy was evaluated by analysing standard reference materials (Spectrapure Standards SW1 and 2, NIST 1643e, Environment Canada TM 27.3, and TM 61.2) for each analytical session, and the error for each element was $<$15%.

3. Results and discussion

3.1. ICP-OES and ICP-MS

All the elements obtained by ICP-OES and ICP-MS techniques are: Cu, Zn, Mn, Fe, Ni, Co, As, Ag, Cd, Sn, Sb, Te, Bi, Pb, Li, Al, V, Cr, Rb, Sr, Ba. Their amounts are summarized in Table 1.

The recovery, in some cases, is substantially different from 100 because of the errors due to the low quantity of sample available. This discrepancy is most significant for samples S38 and S08, whose quantities were 0.0123 and 0.0160 g, respectively.

The concentration of the main elements Cu, Zn and Pb ranges from 69 to 90%, from 16 to 28% and from 1.0 to 7.7%, respectively. It is important to underline that, as it was reported in the previous paper [2], the highest zinc concentration is 28% suggesting that the ingot production is compatible with the cementation process [16].

As already observed by XRF analysis [2], the variability of Cu/Zn (w/w) ratio indicates that the manufacturer had not a preferential value of composition, or a reliable way to control it; alternatively, the ingots came from different furnaces, possibly using raw minerals from different geographical areas.

It is important to highlight that the concentration of Pb, ranging from 1.0% to 7.7% indicates, at least for the highest concentrations, a voluntary addition of the metal. This addition is useful when a large casting as in statues is needed; the brass so obtained has poor mechanical features but a lower melting point.

A rapid inspection of the data in the table, however, immediately shows three outlier samples: S13, S25 and S38. With respect to the other samples, they have a low Pb content [S13 (3.30%), S25 (1.05%) and S38 (2.33%)], furthermore the samples S13 and S38 have high As, Mn and Bi content [As (ca 0.02%), Mn (0.008–0.009%) and Bi (0.01–0.02%)]. Sample S13 has a significantly higher amount of Sb (0.05%).

The different trace elements or the difference in their concentration are probably due to different metal minerals accompanying raw materials or to their introduction by handling methodologies, therefore the adequate analysis of the trace elements could be useful in pinpointing the provenance of the materials or technological differences. As a consequence, in order to get an insight on these precious findings the multivariate analysis of the data was applied.

3.2. Multivariate treatment

There is a large variability of the content of each element through the samples and even a remarkable difference of the amounts of different elements in the same sample. In order to overcome this problem we select the Median Absolute Deviation (MAD) scaling to normalize scale and offset. For this purpose the data were arranged in a matrix containing in each column the concentration values of an element over all samples (hereafter named variable) while each row store the elemental composition of an ingot. MAD is defined as the median of absolute differences between each observation in the column and the median observation. The MAD scaling here applied is a subtraction of the column median value followed by division by the column MAD value multiplied by the factor 1.4826.

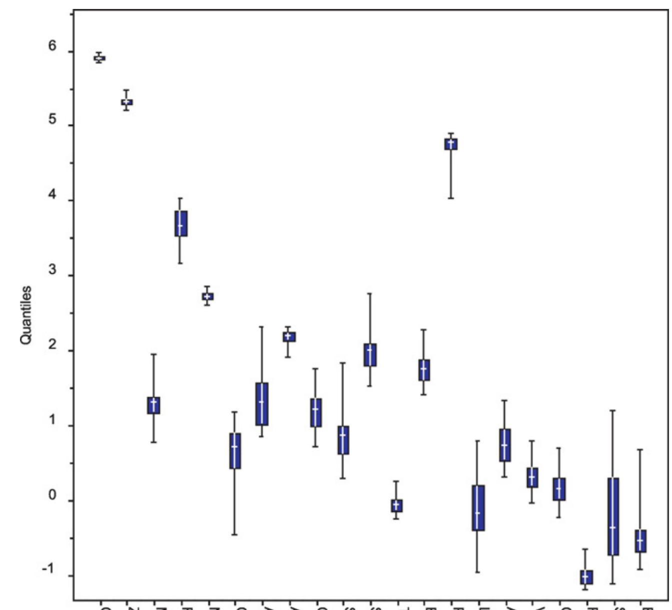

Fig. 3. Plot of the logarithmic transform of the variables. Here the blue box span from 25% to 75% percentiles, the central line mark the median and the whiskers show the minimum and the maximum values.

The MAD multiplied by the factor makes it a robust estimator of the standard deviation when many observations are collected from a normal distribution. This kind of scaling permits to highlight the presence of outlying samples.

Our data, furthermore, include main elements at high concentration and trace elements with very low concentration values so that a logarithmic transform before the normalization improves the result scaling.

It can be observed, in Fig. 3, that most of the elements have a large dispersion with some of them more evident and some other with unidirectional spikes.

3.3. Clustering

Since no clear trend is evident in the multivariate treatment, it was attempted to disclose by cluster analysis, if some grouping is justified. The trial of several hierarchical methods produces, for most of them, similar results. The dendrogram obtained by the Ward's method is reported in Fig. 4.

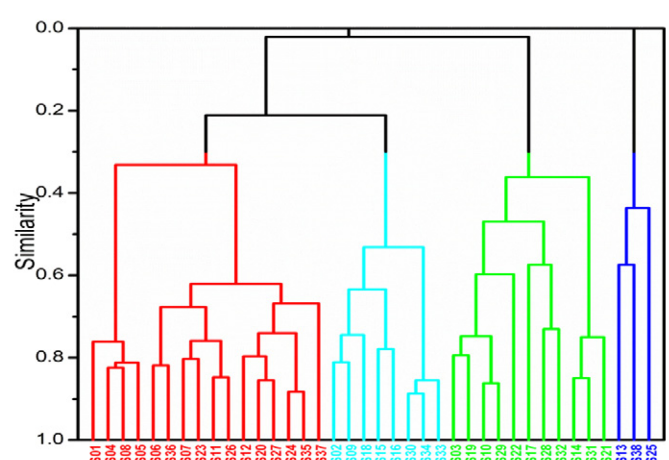

Fig. 4. Clustering with Ward method by PARVUS.

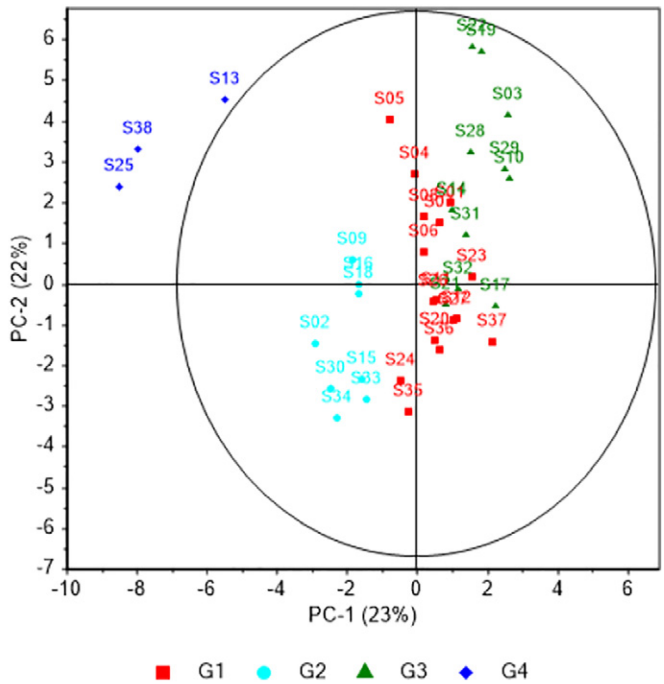

Fig. 5. Projection PC-1 PC-2 of the log transformed and MAD scaled data. Colors define the groups obtained by Ward cluster method after MAD scaling plus auto-scaling of the data.

Data cluster fall into three main groups and a further one formed by only three ingots. Each of the three main groups has at least two or three subgroups. Applying different clustering techniques two groups remain unchanged, one evidenced by blue lines always well defined, and the other one evidenced by cyan lines, sometimes including samples S17 and S32. The red and green groups are less stable.

In any case, the partitioning of the data by means of supervised methods, LDA and SIMCA was checked hypothesizing the following grouping:

G1: S1, S4, S5, S6, S7, S8, S11, S12, S20, S23, S24, S26, S27, S35, S36, S37 (red group)

G2: S2, S9, S15, S16, S18, S30, S33, S34 (cyan group)

G3: S3, S10, S14, S17, S19, S21, S22, S28, S29, S31, S32 (green group)

G4: S13, S25, S38 (blue group)

The PCA checked this grouping hypothesis; in fact the PCA unsupervised method, permits to verify the grouping hypothesis obtained by the cluster analysis. The projection of the data on the PC-1 (23%) and PC-2 (22%) is reported in Fig. 5.

It is noticeable that even PCA highlights the S13, S25, and S38 out of the 95% confidence reported by the black circle. The colors in the figure were fixed on the base of the colors of the cluster grouping. In this way even PCA shows the sample grouping.

From the findings of these unsupervised method it is interesting to underline that:

i) samples 24 and 35, referring to the tow ingots that are connected together by a sort of metal bridge probably for some spill on the casting process, are located near each other in the red group (G1) of the clustering representation;
ii) ingots S13, S25 and S38 (blue color) are the ones previously indicated as possible outliers. Since they are identified also in cluster analysis and in other trials as a different possible group they were discarded from the following validation because of the reduced number of samples. The three groups G1, G2 and G3 were ones used in the validation procedure with supervised methods.

The distributions of weight, width, length and copper/zinc weight ratio for each of the three main groups are reported in Fig. 6.

Ingots belonging to the group G1, the red one, even if this group is the most numerous, have the most homogeneous distribution of the

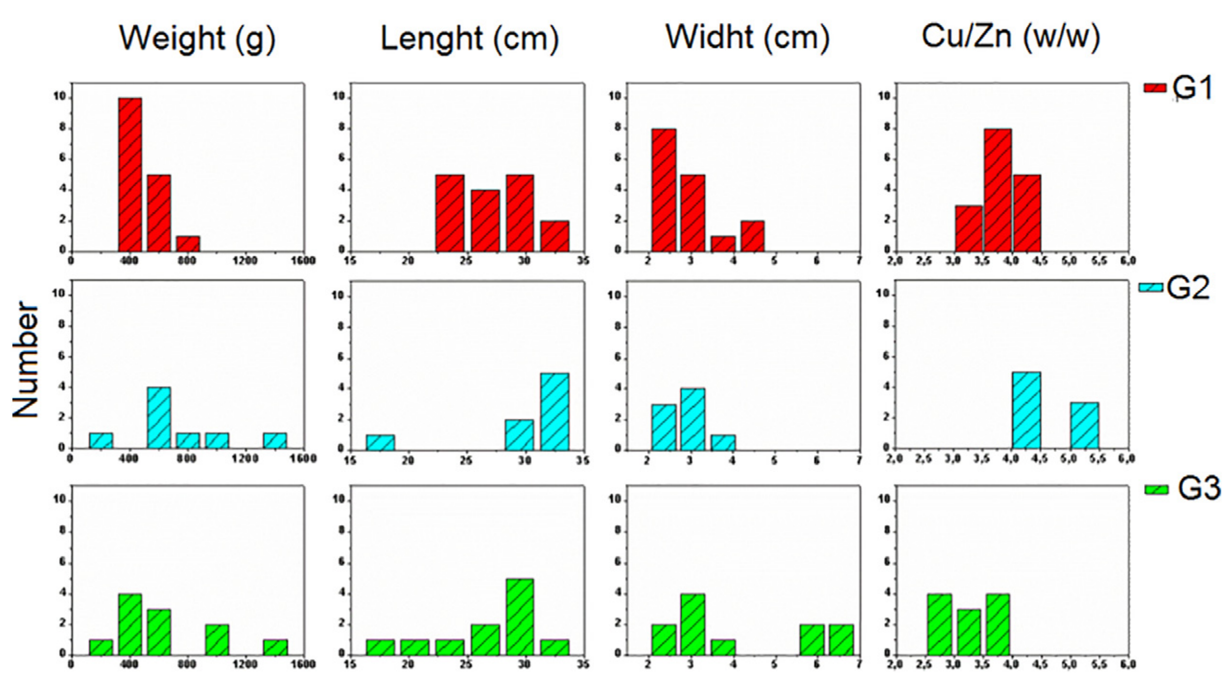

Fig. 6. Weight, length, width and Cu/Zn w/w distributions of G1, G2 and G3 groups.

four analysed parameters. The small differences in the dimensional parameters could indicate that they were produced in a furnace working with a more accurate melting techniques and casting procedures than the others furnaces producing ingots belonging G2 and G3 groups.

The group G2, the cyan one, includes the minor number of ingots (8). One of the ingots is smaller, but probably because it was the last one of the preparation set.

The group G3, the green one, includes two different shapes of ingots, (bimodal distribution of width) the four wider ingots are near each other.

The group G4, the blue one including ingots S13, S38, and S25, is characterized by the most irregular borders and surface.

In order to find some correlations between the results of the clustering analysis and some of the morphological and feature characteristics of the ingots, the pictures of the four groups obtained by the clustering techniques are reported in Fig. 7 following the same order obtained by the clustering techniques.

It is noteworthy the concordance between the classification based on the chemical composition of the ingots with their morphology. This strengthens the evidence of their different provenance because the morphological aspects can be only ascribed to the casting procedure adopted by the furnaces.

It can be interesting to notice that the wider ingots S17, S22, S28 and S32 are positioned near each other in the group G3; their Cu/Zn ratio is similar, while it differs of one unit compared to the first cigar like subgroup. It was claimed [5] that there is a correlation between the temperature reached during the casting procedures and the maximum amount of Zn in the obtained alloy, therefore the observed difference in the Cu/

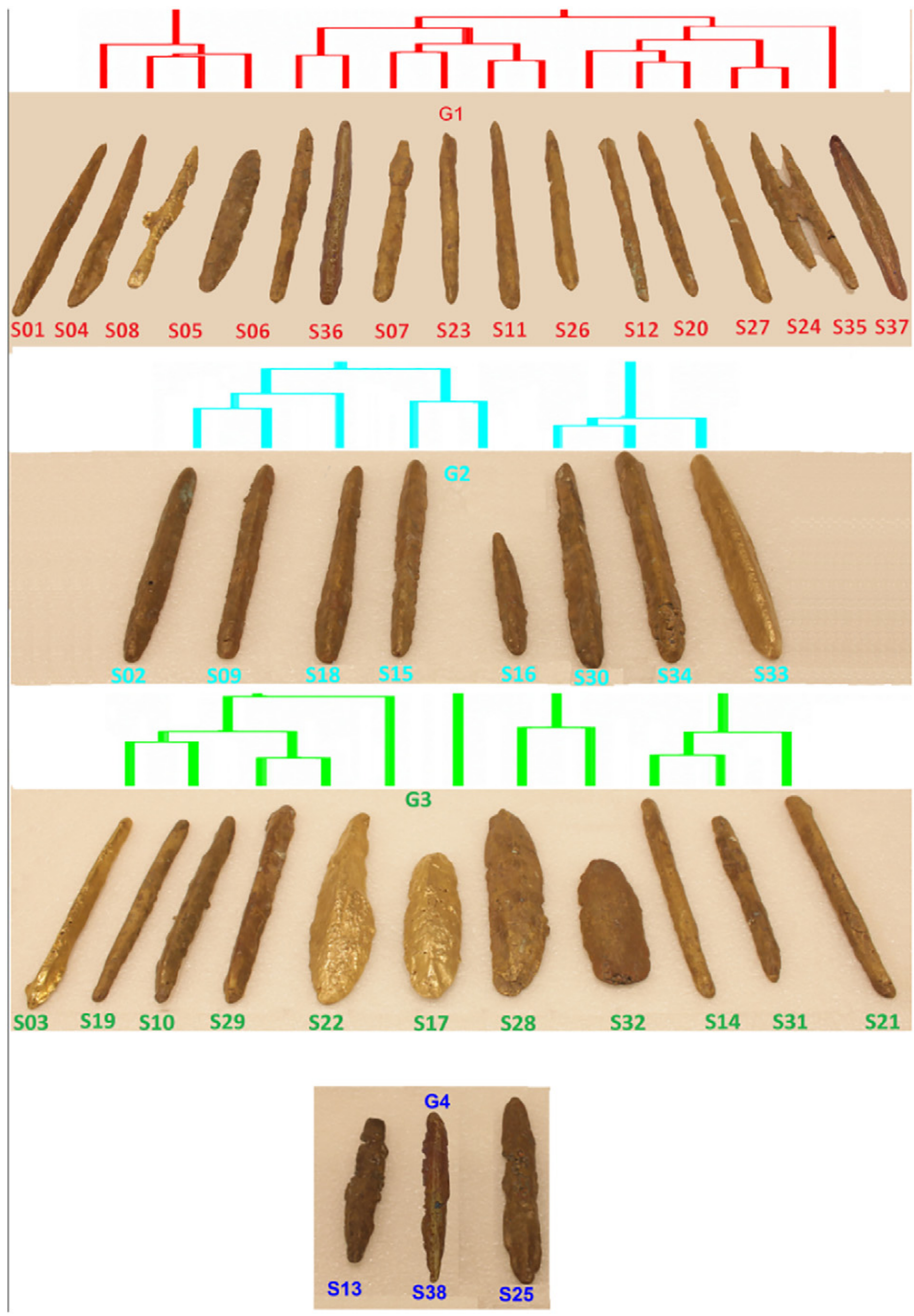

Fig. 7. Photos of the groups obtained by the clustering techniques.

Zn ratio of the two G3 subgroups should indicate that the cigar shape ingots and wider ones were prepared from the same raw materials, but in the first subgroup the crucibles reached higher temperature.

The ingots of the group G4, S13, S25 and S38 while looking similar each other, are very different from ingots of the other groups.

3.4. Supervised validation

The results obtained by treating the data with different supervised pattern recognition methods are described. The ability of the methods to classify the ingots was checked. An estimation of prediction ability, also, is computed by means of leave-one-out validation. Some troubles about the predictively of these methods are due to the poor number of samples however the goal here is simply to confirm with other instruments the grouping found by cluster.

3.5. LDA

The results of LDA method computed using 12 principal components are reported in Table 2. The method recognizes the three groups (TOTAL 100.0% ability) with a comparable prediction ability (TOTAL 91.4% ability).

3.6. SIMCA

SIMCA was performed using 95.0% confidence level for group space, the number of components for the inner space is: 13 for group G1, 5 for group G2 and 8 for group G3.

A very good grouping (TOTAL 100.0% ability) and a nice prediction (TOTAL 91.4% ability) were obtained. Results of SIMCA analysis are reported in Table 3.

These results come with a very high Specificity (100%) (ability of the method to reject samples outside the considered group) for all group even when leave-one-out cross validated. However C.V. sensitivity is very poor and grouping sensitivity has a mean value of 94%. These finding suggest large model for the group that can include even samples from the other groups, this is especially true for groups G1 and G2.

Table 2
Confusion matrix resulting from LDA analysis with 12 principal components. The grouping ability of the model is reported on the left part, while the right part reports the estimated prediction by leave-one-out validation.

		Assigned to group in grouping			Assigned to group when predicted by leave-one-out		
		G1	G2	G3	G1	G2	G3
True group	G1	16	0	0	15	0	1
	G2	0	8	0	0	8	0
	G3	0	0	11	1	1	9
	% correct grouping	100.0	100.0	100.0	93.7	100.0	81.8

Table 3
Confusion matrix resulting from SIMCA analysis. The left part of the table shows grouping ability of the model, while the right part reports the estimated prediction by leave-one-out validation.

		Assigned to group in grouping			Assigned to group when predicted by leave-one-out		
		G1	G2	G3	G1	G2	G3
True group	G1	16	0	0	14	0	2
	G2	0	8	0	1	7	0
	G3	0	0	11	0	0	11
	% correct grouping	100.0	100.0	100.0	87.5	87.5	100.0

Table 4
SIMCA ordered discriminant power of the variables.

Variable	Discriminant power	Variable	Discriminant power
Cd	30.5	Sb	6.1
Ag	17.7	Ba	5.2
Zn	15.1	Ni	4.4
Bi	13.4	Rb	4.2
As	11.7	Cu	4.1
Sn	10.8	Al	4.0
Mn	9.6	V	4.0
Fe	8.7	Li	3.4
Pb	8.6	Sr	3.3
Co	8.2	Cr	3.0
Te	7.1		

SIMCA permits, also, to estimate the discriminant power of the variables reported in Table 4.

The goodness of the variable selection made by SIMCA is confirmed plotting the MAD scaled data. As an example, a three-dimensional plot of the first three elements Cd, Ag, and Zn are reported in Fig. 8.

The grouping ability is less than the chemometric methods and fails in few samples but the general behavior is maintained and confirms the goodness of the SIMCA results.

4. Conclusion

The 40 orichalcum ingots discovered near Gela seabed were analysed by using ICP-OES and ICP-MS techniques. The obtained concentrations of the major elements together with the ones of trace elements provided an amount of data useful to perform a statistical analysis by multivariate methods to highlight possible hidden information. Clustering and principal component analyses were used to infer the presence of groups. This approach allowed us asserting the presence of three main groups (G1, G2 and G3) plus three samples that both PCA and Ward's clustering analysis suggested as outlier samples. A further confirmation of the three main groups was obtained with supervised

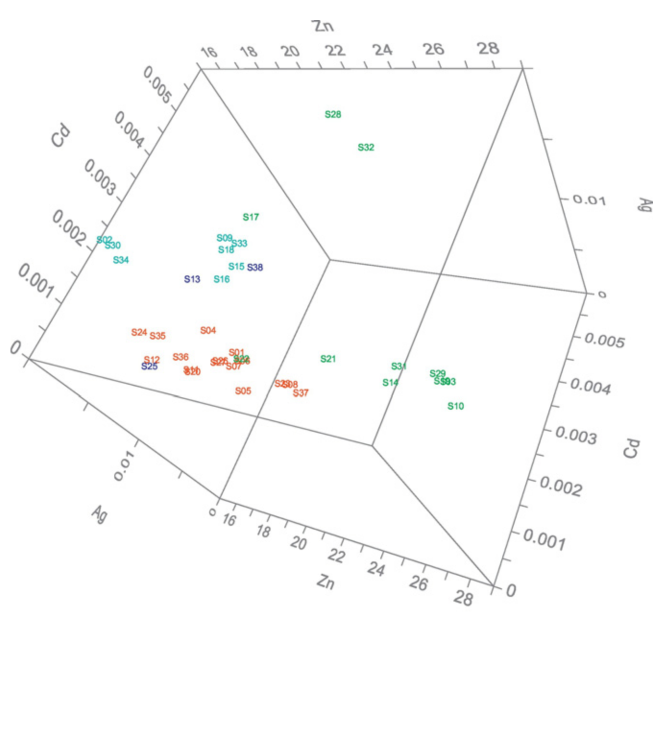

Fig. 8. Plot of samples, including S13, S25, and S38, in the space of the first three SIMCA most discriminant variables. Values are in percentage.

methods, LDA and SIMCA, which show a high grouping ability and even a good prediction ability estimated with leave-one-out validation.

The presence of the three well-defined groups could be ascribed to different manufactures in the production methods and/or in the provenience of the minerals as the elements with the larger discriminant power are not the main component of the alloy but some of the impurities. It is noteworthy the concordance between the classification based on the chemical composition of the ingots with their morphology. This strengthens the evidence of the different origins of the ingots because the morphological aspects can be only ascribed to the casting procedure adopted by the furnaces.

This study represent the base of a deep investigation on the orichalcum ingots, aimed to obtain more information about the production and the origin of these interesting artefacts and the Mediterranean trade in ancient Greek times. Furthermore these results must be framed within the archaeological context in order to hypothesize the location of the furnaces. On this basis, it deserves to be stressed that obtained results are in line with the archaeological chronology given by the context.

It is worth to point out that the results of the above described analysis are not exhaustive but they are of outmost importance because on one side they are not in contrast with the archaeological chronology given by the context, and on the other constitute the base to perform further investigation aimed to the identification of possible sources of the metals used to produce orichalcum as well as the way of manufacturing them.

It is early to make hypothesis, but we intend to proceed towards the deepening of researches aimed to verify if in Anatolia it is possible to find ores and/or workshops where such ingots were produced. This geographical area has been chosen based on the chronological frame of our ingots, around the VI century BC, and could be connected to the rich production of orichalcum objects in some important towns of western Anatolia such as Gordion. Moreover we have to remember that in some ancient town of Phrygia, a little bit later (in Hellenistic and Roman periods) there was a quite wide production of orichalcum objects, mainly coins, indicating that this alloy and the composing minerals were known and present in that area.

Furthermore we have to take in mind that the context of the cargo in which our ingots traveled contains objects coming from eastern Mediterranean and Aegean.

Acknowledgements

This work is part of the project "Development and Application of Innovative Materials and processes for the diagnosis and restoration of Cultural Heritage - DELIAS" - PON03PE_00214_2 (Programma Operativo Nazionale Ricerca e Competitività 2007–2013).

Thanks are due to Museo Archeologico Regionale di Gela, Gela (CL); Capitaneria di Porto di Gela, Gela (CL); Comando Provinciale Guardia di Finanza Caltanissetta; Stazione Navale Palermo-Guardia di Finanza, Palermo to have supported our research making available the ingots.

Thank are due to the discoverer Mr. Francesco Cassarino. We are thankful to Prof. Vincenzo Gennaro and to Mr. Riccardo Cassinera for the useful discussion and suggestions.

References

[1] D. Vullo, La nave greca arcaica di Gela, Dallo scavo al recupero BetaGamma Editoria 4, 2013 (256 pp.).
[2] E. Caponetti, F. Armetta, D. Chillura Martino, M.L. Saladino, S. Ridolfi, G. Chirco, M. Berrettoni, P. Conti, B. Nicolò, S. Tusa, First discovery of orichalcum ingots from the remains of a 6th century BC shipwreck near Gela (Sicily) seabed, Mediterranean Archaeology and Archaeometry 17 (2) (2017) 11–18.
[3] A.K. Biswas, The primacy of Indi in ancient brass and zinc metallurgy, Indian J. Hist. Sci. 28 (4) (1993) 309–330.
[4] E. Caley, Orichalcum and Related Ancient Alloys, 13–31, American Numismatic Soc., New York, 1964 13–31.
[5] X. Fan, G. Harbottle, Q. Gao, W. Zhou, Q. Gong, H. Wang, X. Yug, C. Wang, Brass before bronze? Early copper-alloy metallurgy in China, J. Anal. At. Spectrom. 27 (2012) 821–826.
[6] A.G. Karydas, Application of a portable XRF spectrometer for the non-invasive analysis of museum metal artefacts, Ann. Chim. 97 (2007) 419–432.
[7] M.F. Guerra, Elemental analysis of coins and glasses, Appl. Radiat. Isot. 46 (6–7) (1995) 583–588.
[8] A. Jurado-López, M.D. Luque De Castro, Chemometric approach to laser-induced breakdown analysis of gold alloys, Appl. Spectrosc. 57 (3) (2003) 349–352.
[9] F.R. Doucet, T.F. Belliveau, J.L. Fortier, J. Hubert, Use of chemometrics and laser-induced breakdown spectroscopy for quantitative analysis of major and minor elements in aluminum alloys, Appl. Spectrosc. 61 (3) (2007) 327–332.
[10] P. Christopher, J. Archaeol. Sci. 29 (2002) 1451–1460.
[11] A.T. Machado, J.R. Matos, F.M.S. Carvalho, A.A.S. Araujo, M.G. Silva-Valenzuela, Characterization of steel production dust and their use in structural ceramics chapter characterization of minerals, metals, and materials, The Minerals, Metals & Materials Series, 2017, pp. 593–600.
[12] P. Zhou, M.J. Hutchison, J.W. Erning, J.R. Scully, K. Ogle, An in situ kinetic study of brass dezincification and corrosion, Electrochim. Acta 229 (2017) 141–154.
[13] O. Oudbashi, M. Hessari, A. Hasanpour, A. Shishegar, A multianalytical approach for the study on manufacturing process in ancient bronze coffin from Luristan, Western Iran, Materials Performance and Characterization 6 (3) (2017) 209–223.
[14] F. Di Turo, N. Montoya, J. Piquero-Cilla, C. De Vitoa, F. Coletti, G. Favero, A. Doménech-Carbó, Archaeometric analysis of Roman bronze coins from the Magna Mater temple using solid-state voltammetry and electrochemical impedance spectroscopy, Anal. Chim. Acta 955 (22) (2017) 36–47.
[15] O. Oudbashi, A. Hasanpour, Bronze alloy production during the Iron Age of Luristan: a multianalytical study on recently discovered bronze objects, Archaeol. Anthropol. Sci. 9 (37) (2017) 1–16.
[16] D. Bourgarit, F. Bauchau, The ancient brass cementation processes revisited by extensive experimental simulation, Feature Archaeotechnology 62 (3) (2010) 27–33.